

MFE 230GA · FINAL PROJECT · FALL 2026

Follow the Workers: Results Book

Every result, figure, table and claim behind the report, with its source file

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MFE 230GA – Active Equity Management

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1 Verdict

VERDICT.md is written at Stage 3.

2 Scoreboard

IC is one-month except V2d (12-month, its holding horizon). DSR is the research-period value with 41 nominal trials for the frozen studies and the test-window value with 112 trials for v2. Labels: conf. = confirmatory, expl. = exploratory follow-up. Source: [results/scoreboard.csv](#).

Strategy	Label	Res.	Test	Full	18m	IC (t)	α/m (t)	Plac.	DSR	Verdict
A0 fixed rule (Sonnet)	conf.	-0.44	-0.36	-0.48	-1.56	-0.020 (-0.71)	-0.25% (-0.92)		0.00	negative, not primary
A1 ridge (Sonnet)	conf.	-0.54	-0.51	-0.52	+0.73	-0.014 (-0.74)	-0.35% (-1.72)		0.00	negative, not primary
A1-T tightness (Sonnet)	conf.	-0.55	-0.59	-0.58	-0.37	-0.029 (-1.42)	-0.42% (-2.06)		0.00	negative, not primary
A3 primary (Sonnet)	conf.	-0.13	-0.58	-0.44	+0.85	-0.021 (-1.08)	-0.38% (-1.86)	0.72	0.00	Do not implement
A4 islands (Sonnet)	conf.	-0.18	-0.79	-0.56	+1.23	-0.020 (-0.86)	-0.55% (-2.39)		0.00	negative, not primary
A3 (Opus follow-up)	expl.	-0.48	-0.46	-0.46	-0.64	+0.001 (+0.05)	-0.24% (-1.54)		0.00	exploratory, negative
A4 primary (Opus follow-up)	expl.	-0.40	+0.01	-0.13	-1.41	-0.002 (-0.10)	+0.01% (+0.05)	0.56	0.00	exploratory, fails G2-G5
V2a wage growth (headline)	post-hoc	+0.09	+0.16	+0.15	-1.71	+0.049 (+2.04)	+0.06% (+0.29)	0.02	0.02	noise after the looks taken
V2b signed composite	post-hoc	-0.06	-0.15	-0.11	-1.34	+0.042 (+1.65)	-0.18% (-0.84)	0.05	0.00	noise after the looks taken
V2c wage + momentum	post-hoc	+0.27	+0.35	+0.32	-0.78	+0.032 (+1.21)	+0.22% (+1.36)	0.26	0.09	noise after the looks taken
V2d -(hires+quits), 12m	post-hoc	+0.50	+0.29	+0.35	+0.29	+0.096 (+2.43)	+0.26% (+1.37)	0.85	0.06	noise after the looks taken

3 Figures

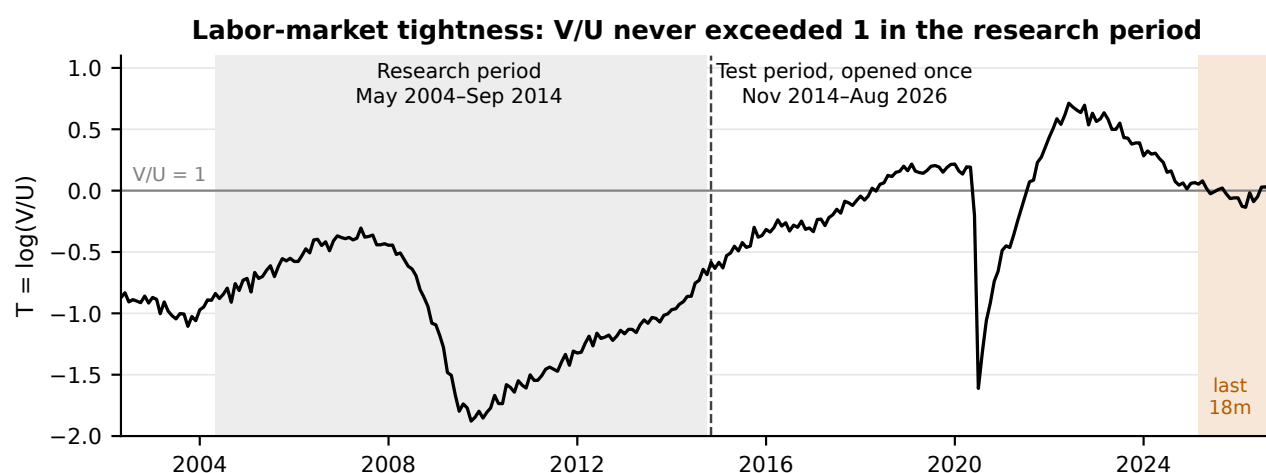


FIGURE 1. Tightness never exceeded $V/U = 1$ in the research period, so the tightness moderator could not be learned there. Source: [analysis/data/fred](#) (JTSJOL, UNEMPLOY). POST-HOC

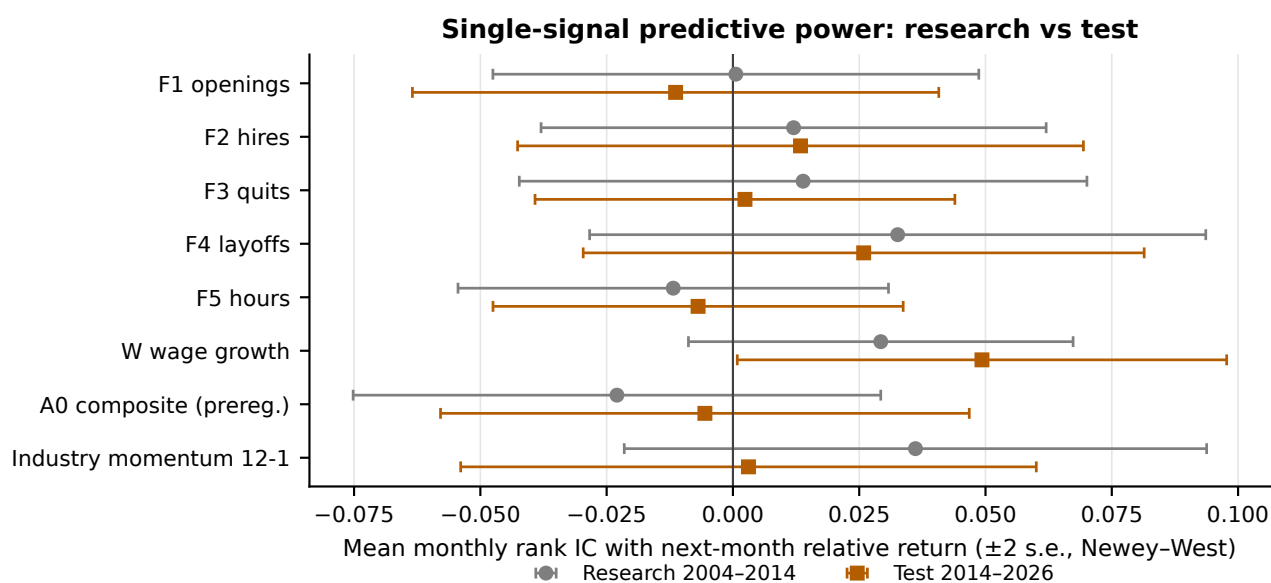


FIGURE 2. Only wage growth has a positive IC in both periods; layoffs and hours carry the opposite sign to the one A0 imposed. Source: analysis/output/tables/d1_signal_ic.csv. POST-HOC

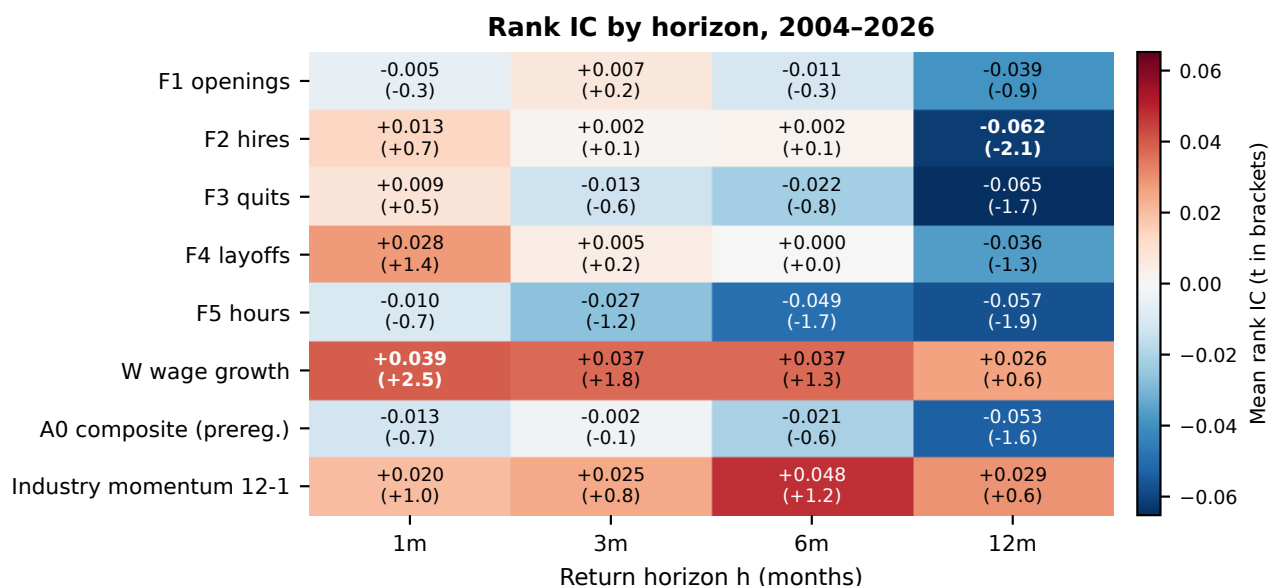


FIGURE 3. Hires and quits turn negative at 12 months (t -2.1, -1.7): the investment/cost channel shows up slowly, not at one month. Source: analysis/output/tables/d2_horizon_ic.csv. POST-HOC

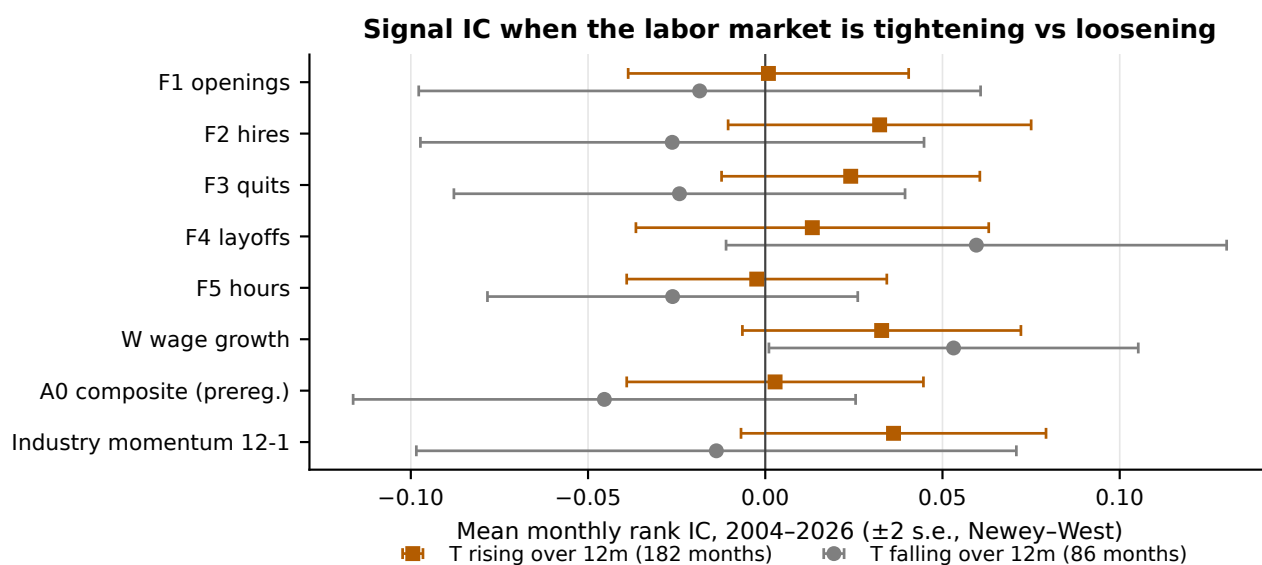


FIGURE 4. Split by whether T rose or fell over 12 months, hires and quits predict returns only while the market tightens; the split is balanced in both periods. Source: analysis/output/tables/d3_tightness_ic.csv. POST-HOC

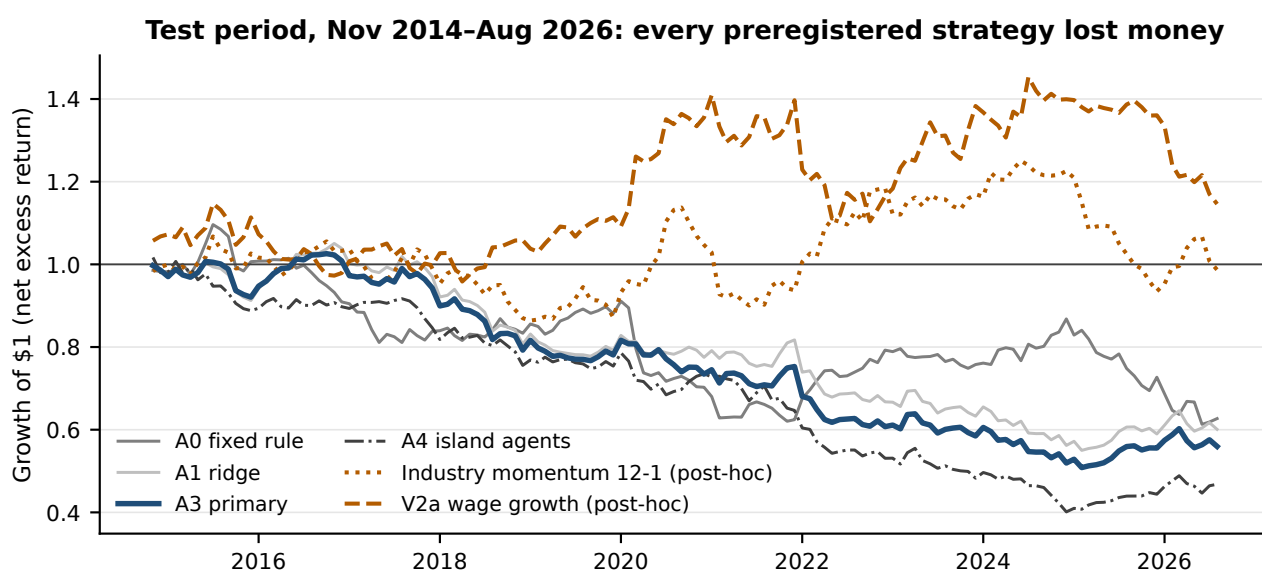


FIGURE 5. Every preregistered strategy lost money in the test; the post-hoc wage-growth book gained until 2024 and gave much of it back. Source: sealed/initial/ASOF_*_ledger.csv; analysis/output/tables/v2_monthly_net.csv. CONFIRMATORY (A0-A4) / POST-HOC (momentum, V2a)

A3 test-period P&L by indus

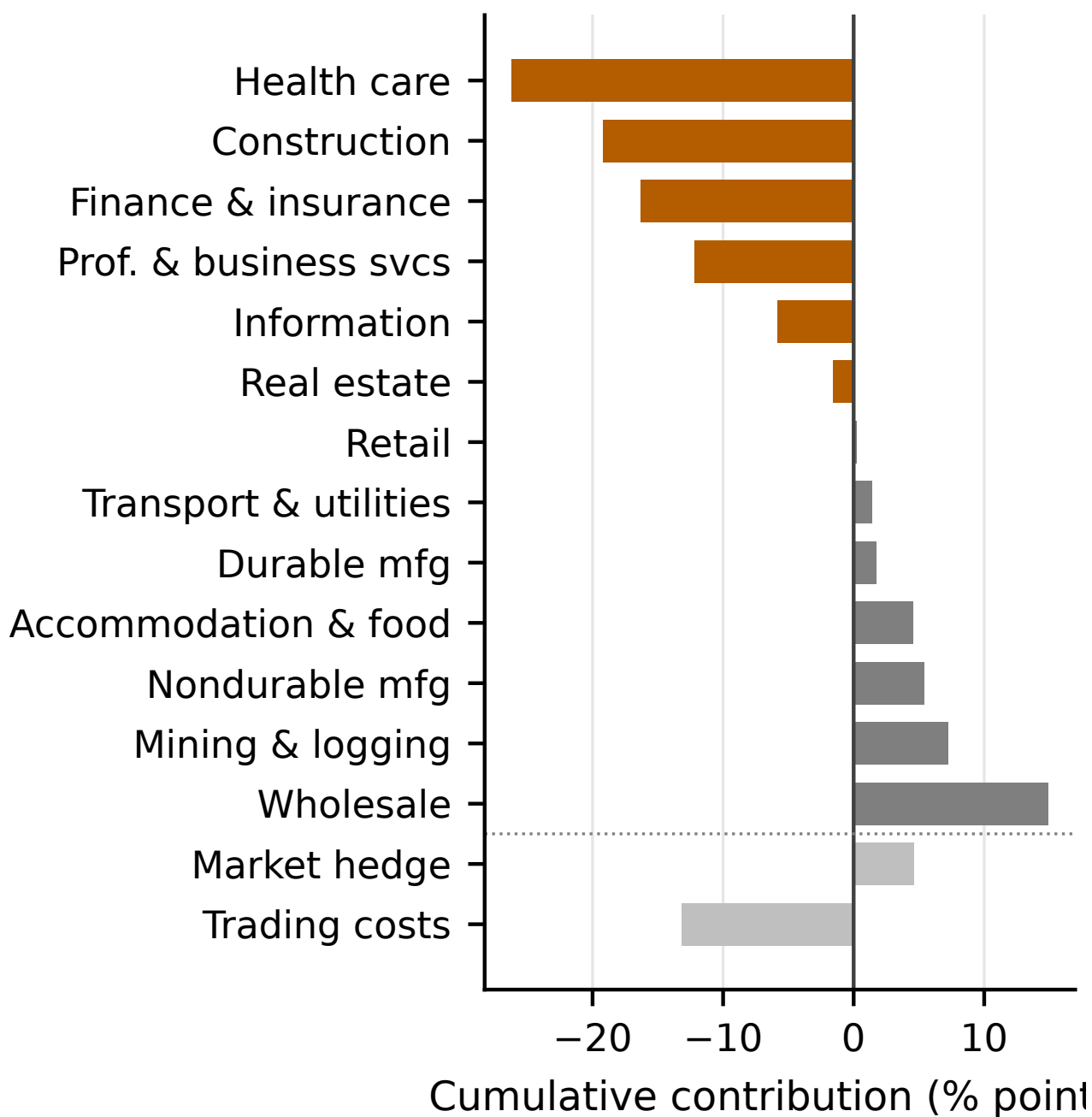


FIGURE 6. A3 lost most in health care, construction and finance; trading costs were a secondary drag. Source: analysis/output/tables/d5_industry_pnl.csv.

POST-HOC

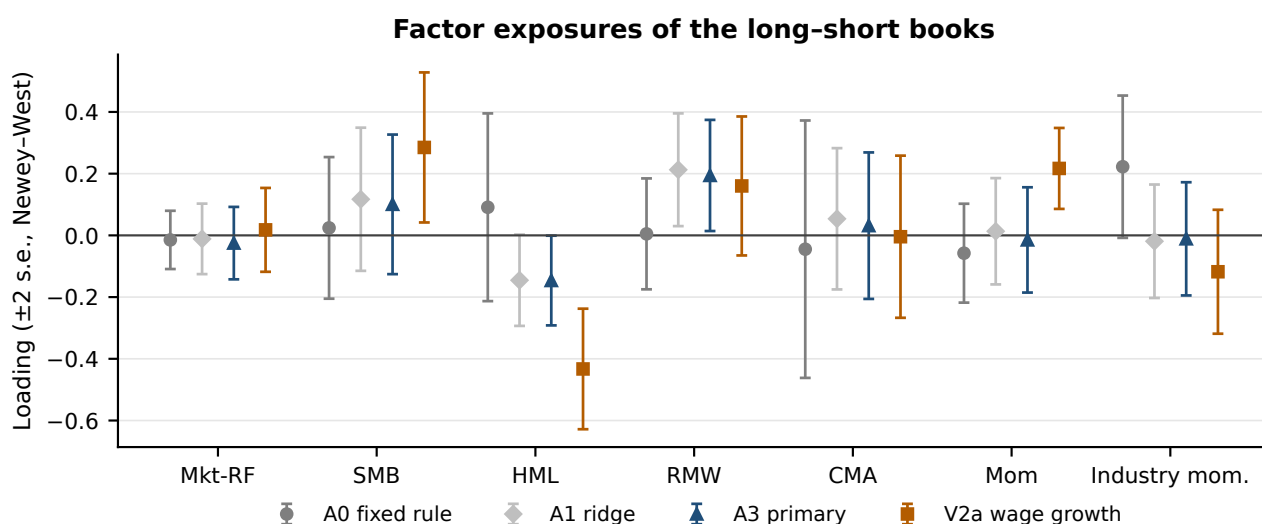


FIGURE 7. The labor books are quality/growth tilts: A3 loads on RMW and against HML; V2a loads against HML and on momentum. Source: results.json attribution; analysis/output/tables/v2a_factor_loadings_test.csv.

CONFIRMATORY (A0, A1, A3) / POST-HOC (V2a)

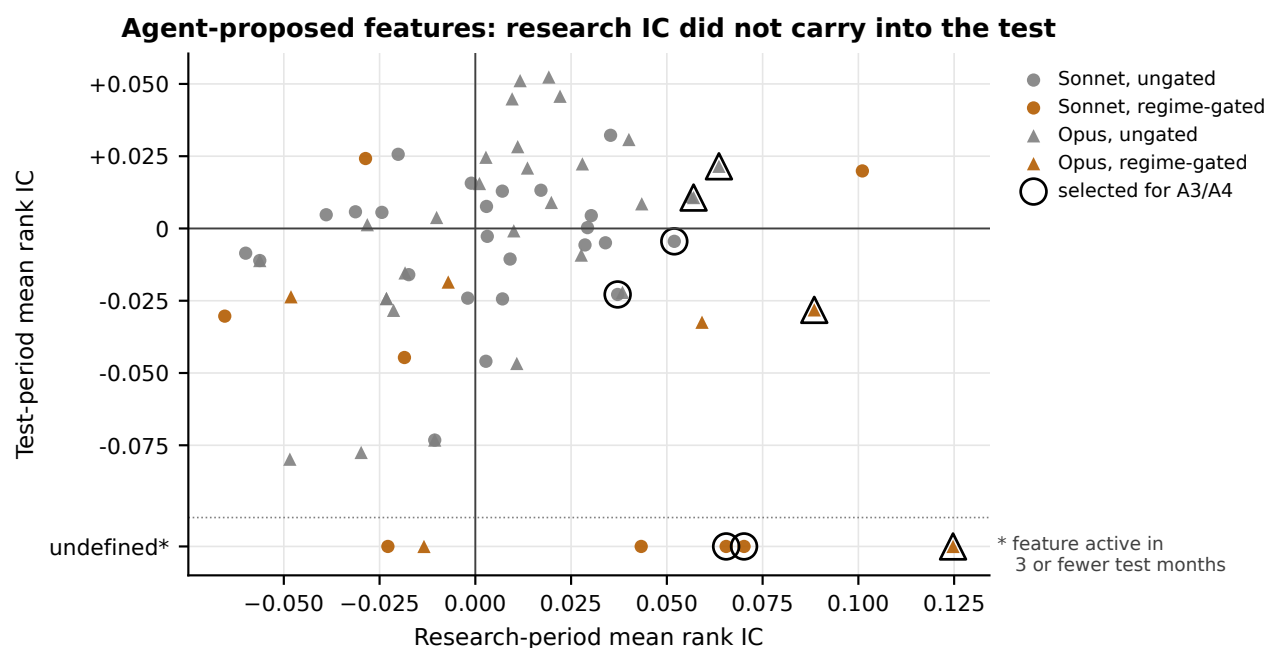


FIGURE 8. Research IC of agent features barely predicts their test IC; the selected regime-gated features were active in 3 test months. Source: analysis/output/tables/d7_agent_candidates.csv.

POST-HOC

Blinded agent experiment: two arms with equal budgets

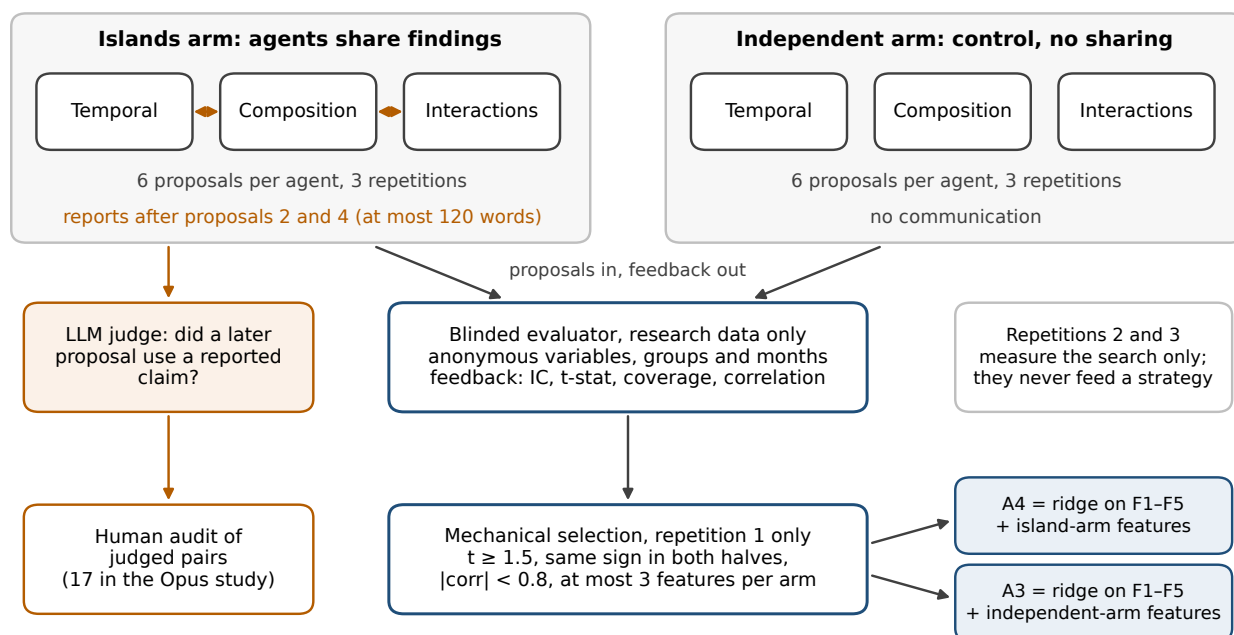


FIGURE 9. Two arms with equal budgets, a blinded evaluator, mechanical selection and a human-audited LLM judge. Source: frozen params.py AGENT. CONFIRMATORY

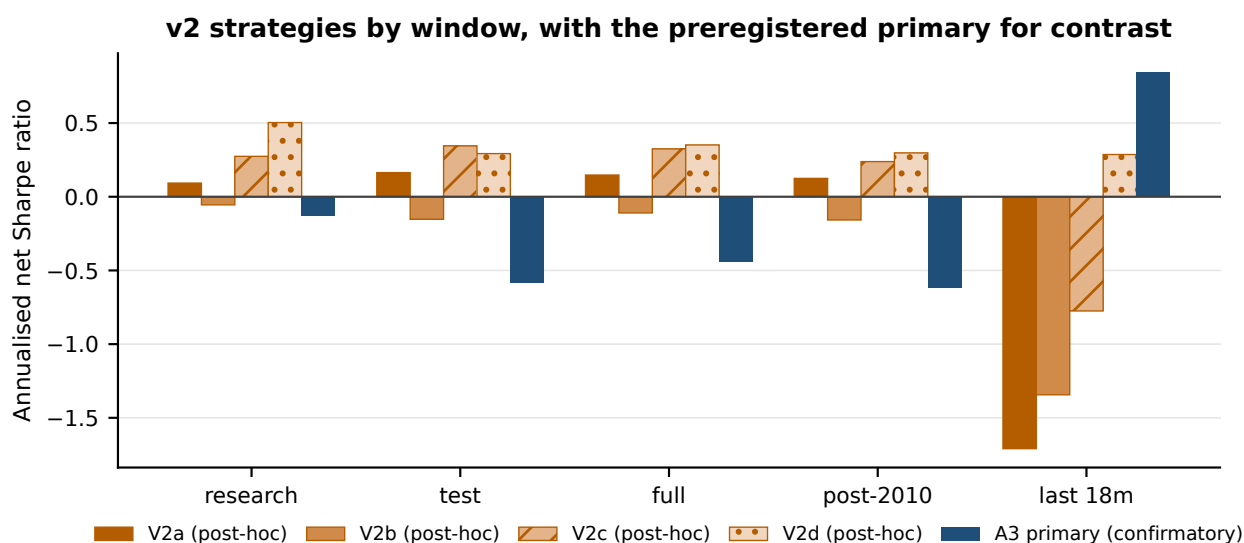


FIGURE 10. V2d is the only variant positive in every window; V2a and V2c lost heavily in the last 18 months. Source: analysis/output/tables/v2_window_stats.csv. POST-HOC / CONFIRMATORY (A3 bars)

4 Statistics

Each table lives in `analysis/output/tables/<name>.csv`; the LaTeX fragment in `report/tables/<name>.tex`.

D1 single-signal rank IC

post-hoc

Signal	Research 2004–14	Test 2014–26	Full 2004–26
F1 openings	+0.001 (+0.02)	−0.011 (−0.44)	−0.005 (−0.30)
F2 hires	+0.012 (+0.48)	+0.013 (+0.48)	+0.013 (+0.71)
F3 quits	+0.014 (+0.49)	+0.002 (+0.11)	+0.009 (+0.50)
F4 layoffs	+0.033 (+1.07)	+0.026 (+0.93)	+0.028 (+1.38)
F5 hours	−0.012 (−0.55)	−0.007 (−0.34)	−0.010 (−0.67)
W wage growth	+0.029 (+1.54)	+0.049 (+2.04)	+0.039 (+2.50)
A0 composite (prereg.)	−0.023 (−0.88)	−0.006 (−0.21)	−0.013 (−0.69)
Industry momentum 12-1	+0.036 (+1.25)	+0.003 (+0.11)	+0.020 (+0.99)

D2 IC by horizon

post-hoc

Signal	1m	3m	6m	12m
F1 openings	−0.005 (−0.31)	+0.007 (+0.23)	−0.011 (−0.28)	−0.039 (−0.92)
F2 hires	+0.013 (+0.70)	+0.002 (+0.07)	+0.002 (+0.08)	−0.062 (−2.08)
F3 quits	+0.009 (+0.50)	−0.013 (−0.55)	−0.022 (−0.78)	−0.065 (−1.73)
F4 layoffs	+0.028 (+1.37)	+0.005 (+0.17)	+0.000 (+0.01)	−0.036 (−1.28)
F5 hours	−0.010 (−0.67)	−0.027 (−1.24)	−0.049 (−1.69)	−0.057 (−1.86)
W wage growth	+0.039 (+2.51)	+0.037 (+1.77)	+0.037 (+1.28)	+0.026 (+0.63)
A0 composite (prereg.)	−0.013 (−0.69)	−0.002 (−0.09)	−0.021 (−0.61)	−0.053 (−1.58)
Industry momentum 12-1	+0.020 (+0.99)	+0.025 (+0.80)	+0.048 (+1.24)	+0.029 (+0.63)

D3a tightness regime balance

post-hoc

Months in regime	Research 2004–14	Test 2014–26
months	125	142
expanding low	49	3
expanding middle	26	6
expanding high	50	133
T rising (12m)	93	88
T falling (12m)	32	54
V/U > 1	0	75
T min	−1.88	−1.61
T max	−0.30	+0.71

D3b IC when T rises vs falls

post-hoc

Signal	T rising (182 m)	T falling (86 m)	Difference
F1 openings	+0.001 (+0.05)	−0.019 (−0.47)	+0.019
F2 hires	+0.032 (+1.51)	−0.026 (−0.74)	+0.059
F3 quits	+0.024 (+1.32)	−0.024 (−0.76)	+0.048
F4 layoffs	+0.013 (+0.53)	+0.060 (+1.69)	−0.046
F5 hours	−0.002 (−0.13)	−0.026 (−1.00)	+0.024
W wage growth	+0.033 (+1.67)	+0.053 (+2.04)	−0.020
A0 composite (prereg.)	+0.003 (+0.13)	−0.045 (−1.28)	+0.048
Industry momentum 12-1	+0.036 (+1.68)	−0.014 (−0.33)	+0.050

D4 test-period performance

confirmatory (A0-A4) / post-hoc (momentum)

strategy	Sharpe	Net p.a.	Vol.	Max DD	Turnover	Costs p.a.
A0 fixed rule	−0.36	−3.5%	9.7%	−40.3%	0.83	1.01%
A1 ridge	−0.51	−4.0%	7.9%	−37.7%	0.91	1.11%
A3 primary	−0.58	−4.6%	7.8%	−41.1%	0.91	1.11%
A4 island agents	−0.79	−6.1%	7.8%	−53.3%	1.11	1.35%
Industry momentum 12-1	+0.03	+0.3%	9.4%	−20.8%	0.53	0.65%

D5 A3 P&L by industry

post-hoc

component	Contribution (pp)	Months long	Months short
Mining & logging	+7.2	33%	36%
Construction	−19.2	36%	39%
Durable mfg	+1.8	31%	24%
Nondurable mfg	+5.4	32%	35%
Wholesale	+14.9	31%	42%
Retail	+0.2	39%	35%
Transport & utilities	+1.4	47%	30%
Information	−5.8	30%	37%
Finance & insurance	−16.3	34%	45%
Real estate	−1.6	24%	45%
Prof. & business svcs	−12.2	18%	47%
Health care	−26.2	41%	36%
Accommodation & food	+4.5	46%	22%
Market hedge	+4.6		
Trading costs	−13.1		

D6 factor loadings

confirmatory

Factor	A0 fixed rule	A1 ridge	A3 primary
Alpha (per month)	−0.25% (−0.92)	−0.35% (−1.72)	−0.38% (−1.86)
Market	−0.01 (−0.31)	−0.01 (−0.20)	−0.03 (−0.43)
SMB	+0.02 (+0.21)	+0.12 (+1.01)	+0.10 (+0.89)
HML	+0.09 (+0.60)	−0.15 (−1.97)	−0.15 (−2.01)
RMW	+0.00 (+0.05)	+0.21 (+2.33)	+0.19 (+2.16)
CMA	−0.04 (−0.21)	+0.05 (+0.47)	+0.03 (+0.26)
Momentum	−0.06 (−0.72)	+0.01 (+0.15)	−0.01 (−0.17)
Industry momentum	+0.22 (+1.93)	−0.02 (−0.21)	−0.01 (−0.12)

D7a agent candidates

post-hoc

Repetition-1 candidates	Sonnet	Opus
valid candidates	33.0	35.0
regime-gated	9.0	6.0
test IC undefined (≤ 3 active months)	4.0	2.0
corr(research IC, test IC)	0.22	0.37
passed research filter	4.0	7.0
selected	4.0	4.0
selected and regime-gated	2.0	2.0
selected with positive test IC	0.0	2.0

D7b selected agent features

post-hoc

Study	Arm	Feature (decoded)	Research IC (n)	Test IC (n)
Sonnet	independent	regime(yoy(Earnings), "low")	+0.065 (48)	n/a (3)
Sonnet	independent	regime(spread(tsz(Layoffs, 60), tsz(Hires, 60)), "low")	+0.070 (43)	n/a (3)
Sonnet	islands	spread(chg(Layoffs, 3), chg(Openings, 3))	+0.037 (123)	−0.023 (142)
Sonnet	islands	spread(chg(Layoffs, 6), chg(Openings, 6))	+0.052 (120)	−0.004 (142)
Opus	independent	mult(regime(tsz(Hires, "exp"), "high"), cstrank(tsz(Openings, "exp")))	+0.088 (15)	−0.028 (133)
Opus	islands	tsz(lchg(lag(Earnings, 1), 12), 36)	+0.064 (78)	+0.022 (142)
Opus	islands	regime(spread(chg(Quits, 3), chg(Hires, 3)), "low")	+0.125 (48)	n/a (3)
Opus	islands	tsz(lchg(lag(Hours, 1), 12), 36)	+0.057 (78)	+0.011 (142)

D8 A3 by window

confirmatory

window	Months	Sharpe	Net return p.a.	Mean IC (t)
Full test, Nov 2014-Aug 2026	142	−0.58	−4.6%	−0.021 (−1.08)
First half	71	−0.72	−4.9%	−0.028 (−0.90)
Second half	71	−0.49	−4.3%	−0.015 (−0.59)
Excluding Mar-Dec 2020	132	−0.51	−4.1%	−0.012 (−0.56)
Last 18 months (Mar 2025-Aug 2026)	18	+0.85	+6.7%	+0.033 (+0.44)
Course window "full" (from Jun 2007)	231	−0.44	−3.2%	−0.009 (−0.49)
Course window "post-2010"	200	−0.62	−4.4%	−0.019 (−1.02)

D9 risk overlay

post-hoc

Study, strategy	Cap binds	Ex-ante vol	Realised vol	Gross leverage	Hedge
Sonnet A0	80%	8.0%	9.7%	1.96	0.15
Sonnet A1	88%	7.5%	7.9%	1.98	0.13
Sonnet A1-T	82%	7.5%	8.2%	1.97	0.12
Sonnet A3	87%	7.5%	7.8%	1.98	0.13
Sonnet A4	91%	7.3%	7.8%	1.99	0.13
Opus A3	92%	7.3%	7.5%	1.99	0.15
Opus A4	93%	7.3%	8.4%	1.99	0.11

D10 agent experiment design

confirmatory

study	item	value
Both	Arms	islands (communicating), independent (control)
Both	Agents per arm	Temporal, Composition, Interactions
Both	Proposals per agent x repetitions	6 x 3
Both	Migration reports	after proposals 2 and 4, at most 120 words
Both	Selection (repetition 1 only)	$t \geq 1.5$, same sign in both halves, $ \text{corr} < 0.8$, at most 3 per arm
Sonnet	Valid proposals	98 / 108
Sonnet	Migration reports admitted	7 / 18
Sonnet	Defined LLM-judge uptake labels	8 / 36
Sonnet	Human-audited judge pairs	8
Opus	Valid proposals	106 / 108
Opus	Migration reports admitted	15 / 18
Opus	Defined LLM-judge uptake labels	84 / 88
Opus	Human-audited judge pairs	17
Opus	Human-judge agreement	10 / 17 (human yes 3, judge yes 10)

v2 variants: Sharpe by window and test statistics

post-hoc

Variant	Research	Test	Full	Post-2010	Last 18m	Test IC (t)	Alpha t	Placebo p	DSR
V2a	+0.09	+0.16	+0.15	+0.12	-1.71	+0.049 (+2.04)	+0.29	0.02	0.02
V2b	-0.06	-0.15	-0.11	-0.16	-1.34	+0.042 (+1.65)	-0.84	0.05	0.00
V2c	+0.27	+0.35	+0.32	+0.24	-0.78	+0.032 (+1.21)	+1.36	0.26	0.09
V2d	+0.50	+0.29	+0.35	+0.30	+0.29	-0.020 (-0.87)	+1.37	0.85	0.06

5 The agent experiment

study	item	value
Both	Arms	islands (communicating), independent (control)
Both	Agents per arm	Temporal, Composition, Interactions
Both	Proposals per agent x repetitions	6 x 3
Both	Migration reports	after proposals 2 and 4, at most 120 words
Both	Selection (repetition 1 only)	$t \geq 1.5$, same sign in both halves, $ \text{corr} < 0.8$, at most 3 per arm
Sonnet	Valid proposals	98 / 108
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Opus	Migration reports admitted	15 / 18
Opus	Defined LLM-judge uptake labels	84 / 88
Opus	Human-audited judge pairs	17
Opus	Human-judge agreement	10 / 17 (human yes 3, judge yes 10)

Selected features and their research→test fate:

Study	Arm	Feature (decoded)	Research IC (n)	Test IC (n)
Sonnet	independent	regime(yoy(Earnings), "low")	+0.065 (48)	n/a (3)
Sonnet	independent	regime(spread(tsz(Layoffs, 60), tsz(Hires, 60)), "low")	+0.070 (43)	n/a (3)
Sonnet	islands	spread(chg(Layoffs, 3), chg(Openings, 3))	+0.037 (123)	−0.023 (142)
Sonnet	islands	spread(chg(Layoffs, 6), chg(Openings, 6))	+0.052 (120)	−0.004 (142)
Opus	independent	mult(regime(tsz(Hires, "exp"), "high"), csrank(tsz(Openings, "exp")))	+0.088 (15)	−0.028 (133)
Opus	islands	tsz(lchg(lag(Earnings, 1), 12), 36)	+0.064 (78)	+0.022 (142)
Opus	islands	regime(spread(chg(Quits, 3), chg(Hires, 3)), "low")	+0.125 (48)	n/a (3)
Opus	islands	tsz(lchg(lag(Hours, 1), 12), 36)	+0.057 (78)	+0.011 (142)

Human vs judge (Opus): 10/17 agreement; human yes 3, judge yes 10; all 7 disagreements judge-yes/human-no. Models from the logs: Sonnet 5.5 at medium effort for the confirmatory study (607 calls), Opus 5.5 at xhigh for the follow-up (905 calls).

6 AI-interaction inventory

ID	Purpose	Model	Prompt file	What was right	What was wrong	What changed	Team critique
P1	Source and code audit before research	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	outputs/follow_the_workers/outputs/p1_request.json	25 numbered findings; found the shared CES series for groups 9/10 and several metadata inconsistencies.	Several alleged timing errors were not supported (release filtering preceeds the month bound); no numeric change resulted.	Mapping meta-data corrected (prescribed vs verified scope); construction SA exception flagged; no basket or feature change.	
P2	Blinded signal-search agents (two arms, 3 repetitions)	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	params.py SYSTEM_PROMPT / TURN_PROMPT; outputs/agents/log.jsonl	Rediscovered wage growth (V05) blind in both studies; produced valid grammar expressions (98/108 Sonnet, 106/108 Opus).	Highest research t-statistics came from regime gates that were active in 3 test months; every selected feature failed the test.	Mechanical selection only; no engineer edits. v2 restores wage growth as a public, ungated signal.	
P3	Migration reports and LLM judge of idea uptake	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	outputs/agents/migrations.json; judge prompt in agents.py trace_migrations	Opus: 15/18 reports admitted and 84/88 usable labels; a human audit of 17 pairs was possible.	Sonnet: 28/36 judge outputs were not bare JSON; judge over-attributes uptake (10 yes vs human 3; all 7 disagreements judge-yes).	Judge output treated as an association, not evidence; human labels reported alongside.	
P4	Robustness red team on research tables	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	outputs/follow_the_workers/outputs/p4_request.json	Ten ranked concerns; called "do not implement" from research tables before the seal opened.	Asserted uncomputed p-values and proposed unregistered thresholds; every suggestion was already covered by a registered diagnostic.	No new test adopted (all dispositions "covered by" or "not adopted"); registered diagnostics retained.	
P5	Post-mortem review of the frozen study	claude-opus-5-5 (Claude Code, 3 Oct 2026)	conversation request (Piero, 3 Oct 2026); FINDINGS_AND_PROPOSAL.md	Found the wrong A0 signs, the dropped wage signal, the 12-month hiring sign, the 3-of-142-month agent features, the one-bin tightness split.	Guessed the engineering assistant was OpenAI Codex (unverified); used a momentum definition that included month m-1; quoted W ICs with a non-frozen start date.	Momentum redefined on m-12..m-2; W ICs recomputed with the frozen feature start; Codex guess withdrawn (to be confirmed by Alex).	
P6	This iteration: rebuild, diagnostics, frozen v2/v3 specs, forward test, (overnight) report	claude-opus-5-5 (Parts 1-3); claude-fable-5-1 (overnight)	ITERATION_PROMPT.md, OVERNIGHT_PROMPT.md, CLAUDE.md	Reproduced the sealed A0 returns to 1e-16; froze specs before running; reported all variants; stopped at a failed gate instead of working around it.	The forward spec assumed August JOLTS would not be out by 29 Sep; it was released that day, so the pre-committed check stopped the run.	Amendment 001 (fixed-lag rule, V2c/V2d added, disclosed); run log guarantees v2 outputs never changed.	

Output files: P1: p1_response.md, p1_review.md; P2: candidates.csv, agents/candidate_records.json; P3: agents/migration_judgments.json, human_review_submission.md; P4: p4_response.md, p4_

review.md; P5: FINDINGS_AND_PROPOSAL.md, analysis/output/diagnostics.md; P6: analysis/run.ipynb, analysis/output/*.md, results/, report/main.tex.

7 Claims register

67 claims, 67 verified. Source: `results/claims.csv`.

ID	Label	Claim	Numbers	Ver.
C01	CONF	The preregistered primary A3 earned a test Sharpe of -0.58 over 142 months (Nov 2014-Aug 2026).	-0.58; 142	y
C02	CONF	A3 lost 4.6% a year net of costs, with 7.8% volatility and a 41% maximum drawdown.	-4.6; 7.8; -41	y
C03	CONF	A3 one-month rank IC in the test was -0.021 (Newey-West $t = -1.08$).	-0.021; -1.08	y
C04	CONF	A3 factor alpha was -0.38% a month after FF5, momentum and industry momentum.	-0.38	y
C04b	CONF	The t-statistic of the A3 alpha was -1.86.	-1.86	y
C05	CONF	The circular-shift placebo p-value for A3 was 0.72.	0.72	y
C06	CONF	A3 break-even trading cost was -32 bp: it loses money even with free industry trading.	-32	y
C07	CONF	A3 passed gate G1 and failed G2, G3, G4 and G5.	True; False; False; False; False	y
C08	CONF	Test Sharpe ratios: A0 -0.36, A1 -0.51, A1-T -0.59, A3 -0.58, A4 -0.79.	-0.36; -0.51; -0.59; -0.58; -0.79	y
C09	CONF	Research-period Sharpe ratios (88 common months): A0 -0.44, A1 -0.54, A1-T -0.55, A3 -0.13, A4 -0.18.	88; -0.44; -0.54; -0.55; -0.13; -0.18	y
C10	CONF	A3 by window: first half -0.72, second half -0.49, excluding Mar-Dec 2020 -0.51, last 18 months +0.85.	-0.72; -0.49; -0.51; 0.85	y
C11	CONF	A3 descriptive course windows: full sample (from Jun 2007) -0.44, post-2010 -0.62.	-0.44; -0.62	y
C12	CONF	A3 Sharpe at 25 bp industry costs was -0.79 and at zero industry costs -0.45.	-0.79; -0.45	y
C13	CONF	A3 loads on profitability (RMW +0.19, $t = 2.2$) and against value (HML -0.15, $t = -2.0$).	0.19; 2.2; -0.15; -2.0	y
C14	CONF	A0 loads +0.22 on industry momentum ($t = 1.9$).	0.22; 1.9	y
C15	CONF	A3 ran at 1.1% a year of trading costs against a gross loss of 3.5% a year.	1.11; -3.5	y
C16	CONF	All four Holm-adjusted comparisons in the Sonnet study had adjusted $p = 1.0$.	1.0; 1.0; 1.0; 1.0	y
C17	CONF	Of the 142 test months, A3 was in the expanding-tercile "high" tightness bin in 133.	133	y
C18	CONF	The cross-validated ridge penalty was 1000, the top of the grid, for every learned model.	1000; 1000; 1000; 1000	y
C19	CONF	Four of four agent features that passed research selection failed the test (Sonnet); six of six in the Opus study.	4; 6	y
X01	CONF	Opus follow-up: primary A4 test Sharpe +0.01, research Sharpe -0.40, placebo $p = 0.56$, Sharpe at 25 bp -0.19.	0.01; -0.4; 0.56; -0.19	y
X02	CONF	Opus A4 one-month IC -0.002 ($t = -0.10$); alpha $t = 0.05$; last 18 months Sharpe -1.41.	-0.002; -0.1; 0.05; -1.41	y
X03	CONF	Opus A3 test Sharpe -0.46.	-0.46	y
X04	POST-HOC	In the Opus A4 book, durable manufacturing alone contributed +0.27 of cumulative gross return in the test.	0.273	y
D01	POST-HOC	Wage growth W: rank IC +0.029 in research ($t = 1.54$), +0.049 in test ($t = 2.04$), +0.039 full sample ($t = 2.50$).	0.029; 1.54; 0.049; 2.04; 0.039; 2.5	y
D02	POST-HOC	Full-sample one-month ICs: openings -0.005, hires +0.013, quits +0.009, layoffs +0.028, hours -0.010, A0 composite -0.013, industry momentum +0.020.	-0.005; 0.013; 0.009; 0.028; -0.01; -0.013; 0.02	y
D03	POST-HOC	Layoffs (F4) carry a positive IC (+0.033 research, +0.026 test) although A0 gave them a negative sign; hours (F5) carry a negative IC (-0.012, -0.007) although A0 gave them a positive sign.	0.033; 0.026; -0.012; -0.007	y
D04	POST-HOC	At a 12-month horizon hires have IC -0.062 ($t = -2.1$) and quits -0.065 ($t = -1.7$); at one month +0.013 and +0.009.	-0.062; -2.1; -0.065; -1.7; 0.013; 0.009	y
D05	POST-HOC	Wage growth IC by horizon: +0.039 at 1 month ($t = 2.5$), +0.026 at 12 months.	0.039; 2.5; 0.026	y
D06	POST-HOC	Expanding tightness terciles put 133 of 142 test months in the "high" bin (3 low, 6 middle); V/U exceeded 1 in 0 research months and 75 test months.	133; 3; 6; 0; 75	y
D07	POST-HOC	A 12-month-change split is balanced: T rising in 93 research and 88 test months, falling in 32 and 54.	93; 88; 32; 54	y
D08	POST-HOC	Industry momentum 12-1 through the same backtest had a test Sharpe of +0.03.	0.03	y
D09	POST-HOC	A3 test P&L by industry: health care -26.2 pp, construction -19.2, finance -16.3, professional services -12.2; wholesale +14.9; trading costs -13.1.	-26.2; -19.2; -16.3; -12.2; 14.9; -13.1	y

ID	Label	Claim	Numbers	Ver.
D10	POST-HOC	Selected agent features: 2 of 4 regime-gated in each study; test IC undefined for 4 Sonnet and 2 Opus candidates; 0 (Sonnet) and 2 (Opus) selected features had a positive test IC.	2; 2; 4; 2; 0; 2	y
D11	POST-HOC	Correlation between research and test IC across valid agent candidates: 0.22 (Sonnet), 0.37 (Opus).	0.22; 0.37	y
D12	POST-HOC	Sonnet's two independent-arm features were active in only 3 of 142 test months; the best Opus island feature (research t 3.56) also in 3.	3; 3; 3.56; 3	y
D13	POST-HOC	The gross-exposure cap bound in 87% of A3 test months; ex-ante volatility averaged 7.5% against a 10% target (realised 7.8%).	87; 7.5; 7.8	y
D14	POST-HOC	Across the preregistered books the cap bound in 80% to 93% of test months.	80; 93	y
A01	CONF	Sonnet study: 98 of 108 proposals valid, 7 of 18 migration reports admitted, 8 of 36 judge labels usable, 8 pairs human-audited.	98 / 108; 7 / 18; 8 / 36; 8	y
A02	CONF	Opus study: 106 of 108 proposals valid, 15 of 18 reports admitted, 84 of 88 judge labels usable, 17 pairs human-audited, human-judge agreement 10 of 17 (human yes 3, judge yes 10).	106 / 108; 15 / 18; 84 / 88; 17; 10 / 17 (human yes 3, judge yes 10)	y
A03	CONF	All 7 human-judge disagreements were judge-yes / human-no (items 2, 3, 8, 12, 13, 16, 17).	7; 0	y
A04	CONF	Judge-labelled uptake of migrated ideas: 25% of usable Sonnet labels (2 of 8), 62% of Opus labels (52 of 84).	0.25; 0.619	y
A05	CONF	Best-so-far research IC, islands vs independent (mean over 3 runs): Sonnet 0.049 vs 0.083, Opus 0.136 vs 0.059; neither gap exceeds the seed-to-seed range.	0.049; 0.083; 0.136; 0.059; False; False	y
V__ V2a__ SR	POST-HOC	V2a net Sharpe (10 bp): research +0.09, test +0.16, full +0.15, post-2010 +0.12, last 18 months -1.71.	0.09; 0.16; 0.15; 0.12; -1.71	y
V__ V2b__ SR	POST-HOC	V2b net Sharpe (10 bp): research -0.06, test -0.15, full -0.11, post-2010 -0.16, last 18 months -1.34.	-0.06; -0.15; -0.11; -0.16; -1.34	y
V__ V2c__ SR	POST-HOC	V2c net Sharpe (10 bp): research +0.27, test +0.35, full +0.32, post-2010 +0.24, last 18 months -0.78.	0.27; 0.35; 0.32; 0.24; -0.78	y
V__ V2d__ SR	POST-HOC	V2d net Sharpe (10 bp): research +0.50, test +0.29, full +0.35, post-2010 +0.30, last 18 months +0.29.	0.5; 0.29; 0.35; 0.3; 0.29	y
V01	POST-HOC	V2a test: one-month IC +0.049 (t 2.04), alpha t 0.29, placebo p 0.02, deflated Sharpe 0.02, Sharpe at 25 bp +0.10, turnover 0.39 per month.	0.049; 2.04; 0.29; 0.02; 0.02; 0.1; 0.39	y
V02	POST-HOC	V2d test: 12-month IC +0.096 (t 2.43), one-month IC -0.020 (t -0.87), alpha t 1.37, pre-declared placebo p 0.85, deflated Sharpe 0.06, Sharpe at 25 bp +0.22.	0.096; 2.43; -0.02; -0.87; 1.37; 0.85; 0.06; 0.22	y
V03	POST-HOC	V2d full sample: alpha t 1.96, 12-month IC +0.071 (t 2.06), deflated Sharpe 0.18, pre-declared placebo p 0.86.	1.96; 0.071; 2.06; 0.18; 0.86	y
V04	POST-HOC	V2d test: net return +2.6% a year, volatility 8.7%, maximum drawdown -14.4%, hit rate 55%, turnover 0.37 per month.	2.6; 8.7; -14.4; 55; 37	y
V05	POST-HOC	V2c test: alpha t 1.36, placebo p 0.26, deflated Sharpe 0.09.	1.36; 0.26; 0.09	y
V06	POST-HOC	V2b test: one-month IC +0.042 (t 1.65) but alpha t -0.84.	0.042; 1.65; -0.84	y
V07	POST-HOC	No v2 variant met the pre-declared reading rule.	not distinguishable from noise after the looks already taken; not distinguishable from noise after the looks already taken; not distinguishable from noise after the looks already taken; not distinguishable from noise after the looks already taken	y
V08	POST-HOC	The highest deflated Sharpe among all v2 variants and windows was 0.18 (V2d, full sample).	0.18	y
V09	POST-HOC	Horizon-matched (12-month IC) placebo for V2d, added post-hoc: p 0.06 test, 0.04 full sample, 0.00 post-2010, 0.26 research.	0.06; 0.04; 0.0; 0.26	y
V10	POST-HOC	V2b signs learned on research data: + for openings, hires, quits, layoffs and wages; - for hours.	1; 1; 1; 1; 1; -1	y
V11	POST-HOC	V2a test loadings: HML -0.43 (t -4.4), momentum +0.22 (t 3.3), SMB +0.28 (t 2.3).	-0.43; -4.4; 0.22; 3.3; 0.28; 2.3	y
F01	FORWARD	V2a Q4 2026 book: long construction, durable manufacturing and information; short mining, health care and accommodation & food.	0.3333333333333333; 0.3333333333333333; 0.3333333333333333; -0.3333333333333333; -0.3333333333333333	y - -

ID	Label	Claim	Numbers	Ver.
F02	FORWARD TEST	V2d Q4 2026 book: long finance, real estate and accommodation & food; short mining, durable manufacturing and retail.	0.3333333333333333; 0.3333333333333333; 0.3333333333333333; 0.3333333333333333; -0.3333333333333333; -0.3333333333333333	y -
F03	FORWARD TEST	SPY hedge per unit tranche: V2a -0.40, V2b -0.06, V2c -0.24, V2d -0.02.	-0.4; -0.06; -0.24; -0.02	y
F04	FORWARD TEST	V2c's long durable manufacturing and short professional services both map to XLI and cancel, so its tradable book has four legs.	0.0	y
F05	FORWARD TEST	The forward spec was frozen at 2026-10-03T09:03:56Z and amendment 001 at 2026-10-03T09:16:11Z, before the book was built at 2026-10-03T09:17:55Z.	True; True	y
S01	POST-HOC	Our rebuilt A0 signal matches the frozen fixed-lag scores with pooled correlation 0.998.	0.998	y
S02	POST-HOC	As-known (ASOF) scores pick the same top three groups as our revised-data signal in only 42% of test months.	42	y
S03	POST-HOC	Durable manufacturing is 60% semiconductors (Chips) by market cap in August 2026; nondurable manufacturing is 60% pharmaceuticals.	Chips 60%, Mach 9%, Autos 7%; Drugs 60%, Hshld 9%, Soda 7%	y
S04	CONF	Gate 1: the 13 group returns match all 50 saved ledgers to 1.5e-16 and our estimators match results.json to 3.6e-15.	1.51e-16; 3.55e-15	y

8 Known issues

- *V2c's ETF legs cancel.* Its long durable manufacturing and short professional services both map to XLI, so the tradable V2c book has four legs (book_2026Q4_etf.csv). Fix or disclosure: Disclose in the forward-test section; track V2c on French group returns as the primary record.
- *Revised vs as-known data.* Every historical v2/v3 number uses revised FRED values with fixed lags (option c in data_vintage.md). The frozen study found the same A3 IC under both rules, but as-known data picks the same top-3 groups in only 42% of months. Fix or disclosure: Label every v2/v3 number "revised data"; R10 (as-known rerun) is not run without a FRED key.
- *Overlap in the 12-month placebo.* The horizon-matched placebo for V2d uses overlapping 12-month returns; the circular shift preserves but does not correct the overlap, so its p-values are optimistic. Fix or disclosure: Keep the pre-declared 1-month placebo ($p = 0.85$) in the reading rule; report the 12-month one as a post-hoc check only.
- *Mapping dilution.* Labor data cover all establishments in an industry; returns cover listed firms. Durable manufacturing is 60% semiconductors and nondurable manufacturing 60% pharmaceuticals by market cap. Fix or disclosure: State as a limit; name NAICS-based CRSP baskets as the follow-up.
- *Power.* 142 test months detect an annualised Sharpe of about 0.6 at $t = 2$. None of the post-hoc Sharpe ratios (0.1-0.35) is distinguishable from zero after deflation. Fix or disclosure: Say so in the limits; the forward test is a record, not a verdict.